Thu Pham

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EDUCATION

University of Pittsburgh — Pittsburgh, PA
Ph.D. Candidate in Oncology Graduate Program — Aug 2024 – Present

Carnegie Mellon University — Pittsburgh, PA
Master of Science, Biotechnology and Pharmaceutical Engineering — Dec 2023

University of Science and Technology of Hanoi — Hanoi, Vietnam
Bachelor of Science, Pharmacological, Medical, and Agronomical Biotechnology — Dec 2020

RESEARCH EXPERIENCE

University of Pittsburgh — Pittsburgh, PA
Advisor: Dr. Wayne Stallaert, Dr. Steffi Oesterreich — May 2025 – Present
Project: Molecular drivers of quiescence in ER+ Invasive Lobular Carcinoma (ILC)

- Deconvolved bulk RNA-seq of ILC tumors (SCAN-B) with InstaPrism using single-cell references to isolate malignant-cell expression, nominating 2 cell line models that best recapitulate tumor cell cycle states for perturbation studies
- Nominated the FOXO3-p27 axis as a candidate driver of ILC quiescence through pathway (GSVA) and transcription factor activity inference, with findings replicated across TCGA, SCAN-B, and METABRIC
- Built an automated quality-control (tissue loss/folding detection) for multiplex immunofluorescence of 800 breast cancer TMA cores and HTAN multiplexed imaging data, enabling single-cell spatial analysis
- Mapped cell cycle phases of individual cells in FFPE breast tumors by combining Gaussian mixture models for cell typing, random forest feature selection, and manifold learning on multiplexed images

Carnegie Mellon University — Pittsburgh, PA
Research Associate; Advisor: Dr. Huaiying Zhang — May 2023 – Jul 2023
Project: The role of overexpressed MAD1 in promoting mitotic slippage in cancer cells

- Optimized purification of a Mad1 coiled-coil (CC)–Mad2-interacting motif (MIM) truncation and characterized its phase separation using affinity chromatography and DIC microscopy
- Automated purification and condition-screening workflows in Emerald Cloud Lab (Mathematica), enabling systematic mapping of phase-separation conditions
- Probed TERRA–SUMOylation protein interactions by pull-down assay and Western blot

University of Science and Technology of Hanoi — Hanoi, Vietnam
Research Assistant; Advisor: Dr. Hai Dang Nguyen — Dec 2020 – Jul 2021

- Evaluated anti-inflammatory activity of plant-derived compounds from *Amomum* species, contributing to 2 publications
- Identified Longifuran A, a novel benzofuran, as an anti-inflammatory candidate with low cytotoxicity and significant inhibition of IL-6, IL-1β, and TNF-α by ELISA and Western blot
- Mentored new students and interns in experimental protocols and managed lab operations, including reagent inventory and equipment scheduling

RELATED COURSEWORK

Introduction to Deep Learning (CMU, 11-785) — Fall 2026
Team project (4): Predicting spatial cell responses with graph neural networks and differentiable diffusion

Bioimaging, Analysis and Spatial Biology (Pitt, MSCBIO 2027) — Spring 2026

Foundations in Cancer Biology (Pitt, MSOGP 2000) — Fall 2024

SKILLS

- **Image Analysis:** Multiplex immunofluorescence (CellDive) processing, illumination/background correction, automated image QC, single-cell feature extraction, cell cycle phenotyping
- **Machine Learning:** PyTorch, scikit-learn; clustering (Gaussian mixture models, Leiden, k-means), Random Forests, manifold learning, dimensionality reduction (PCA, UMAP, NMF)
- **Programming & Tools:** Python (pandas, NumPy, scanpy), R, Go, Git/GitHub
- **Genomics:** Bulk and single-cell RNA-seq, cell type deconvolution, GSVA, TF activity inference; public cohorts (TCGA, SCAN-B, METABRIC, HTAN)
- **Microscopy:** Leica THUNDER, spinning disk confocal, DIC
- **Wet Lab:** Mammalian cell culture, siRNA, FACS, Western blot, ELISA, PCR, cloning; protein purification (affinity chromatography, FPLC), DLS, capillary electrophoresis

LEADERSHIP

Carnegie Mellon University Pittsburgh, PA
Alternate Representative of Master programs in Biological Sciences Department Oct 2022 – Sep 2023

- Advocated for graduate student social programming needs at monthly Graduate Student Assembly meetings, representing departmental interests to faculty and administrative leadership
- Organized quarterly social events for 30+ Master's students, fostering community within the program

TEACHING EXPERIENCE

Carnegie Mellon University Pittsburgh, PA
Graduate Teaching Assistant; Courses: Synthetic Biology, Genome Editing Sep 2023 – Dec 2023

- Provided detailed written feedback on assignments for 40+ students across Synthetic Biology and Genome Editing courses
- Facilitated active learning discussions and group activities, supporting comprehension of advanced topics in gene circuit design and CRISPR-based editing

PUBLICATIONS AND CONFERENCES

- **Pham, A.T.**, Carleton, N., Opsal,A. & Oesterreich, S., Lee, A. Ki-67 and Proliferative Activity in a Grade- and Stage-Matched Analysis of Lobular and Ductal Breast Carcinoma *Under preparation.*
- Yang, Y., Hao, S., Poe, D., **Pham, A.T.**, Brown, D., Foldi, J., Kedziora, K., & Stallaert, W. AMBRA1 downregulation is a mechanism of rapid adaptive resistance to CDK4/6 inhibitors in lobular breast carcinoma. *Preprint, under revision.*
- Le, T.H., **Pham, A.T.**, Phi, T.D., Doan, T.M.H., Pham, V.C., & Nguyen, H.D. (2021). Anti-inflammatory properties of Longifuran A, a new benzofuran from the stems of *Amomum longiligulare*. *Chemistry & Biodiversity.*
- **Pham, A.T.**, Nguyen, H.S., Le, T.H., & Nguyen, H.D. (2021). Anti-inflammatory properties of *Amomum maximum* Roxb. and *Amomum muricarpum* Elmer in the North of Vietnam. *Vietnam Journal of Biotechnology*, 19(2).
- **Conference Abstract:** Yang, Y., **Pham, A.T.**, McLaughlin, J., Oesterreich, S., Lee, A., Kedziora, K., & Stallaert, W. (2025). Cell cycle plasticity drives tumor progression and drug resistance [Conference abstract]. *AACR Annual Meeting*, Abstract SY22-02.

REFERENCES

Wayne Stallaert, Ph.D.
Associate Professor, Department of Computational and Systems Biology
University of Pittsburgh
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Ph.D. Advisor

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Assistant Professor, Department of Biology
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Master's Advisor

Carola Neumann, MD
Associate Professor, Department of Pharmacology and Chemical Biology
University of Pittsburgh
neumannc@upmc.edu
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